

Secure Note-Taking API using ASP.NET Core Web API

Project Objectives

The main objectives of the project are:

- Allow users to register securely.
- Allow users to log in using valid credentials.
- Generate JWT tokens for authenticated users.
- Allow authenticated users to create, view, update, and delete notes.
- Ensure that users can access only their own notes.
- Implement input validation and error handling.

Technologies Used

The following technologies were used:

- ASP.NET Core Web API (.NET 8)
- Entity Framework Core
- SQL Server
- JWT Authentication
- BCrypt Password Hashing
- Visual Studio 2022

Project Flow

The overall workflow of the application is as follows:

1. User Registration
2. A new user creates an account using a username and password.
3. The password is hashed using BCrypt before being stored in the database.
4. User Login
5. The user logs in with valid credentials.
6. The entered password is verified against the stored hashed password.
7. JWT Token Generation
8. After successful login, a JWT token is generated.
9. The token contains user information and is used for authentication.

10. Authorization

11. The user includes the JWT token in API requests.

12. Protected endpoints can only be accessed by authenticated users.

13. Note Management

14. Users can create notes.

15. Users can view their own notes.

16. Users can update existing notes.

17. Users can delete notes.

18. Validation and Error Handling

19. Input data is validated before processing.

20. Appropriate error messages and status codes are returned when necessary.

User Registration

Users can register by providing a username and password.

Registration Process

- Username uniqueness is checked.
- Password is hashed using BCrypt.
- User information is stored securely in the database.

Benefits

- Protects user credentials.
- Prevents storage of plain text passwords.
- Improves overall application security.

User Login

Registered users can log in using their credentials.

Login Process

- Username is verified.
- Password hash is validated using BCrypt.
- JWT token is generated upon successful authentication.

JWT Authentication

JWT authentication is used to secure the API.

Purpose of JWT

- Verifies user identity.
- Protects API endpoints.
- Enables secure communication.
- Eliminates the need for server-side session storage.

Authorization

All note-related endpoints are protected using JWT authentication.

Only authenticated users can access note management features. Each user can access only their own notes, ensuring data privacy and security.

Adding Notes

Authenticated users can create new notes.

Features

- Create notes with title and content.
- Associate notes with the logged-in user.
- Store notes securely in the database.

Viewing Notes

Users can retrieve all notes associated with their account.

Features

- Display user-specific notes.
- Prevent access to notes belonging to other users.
- Return note details in a structured format.

Updating Notes

Users can modify existing notes.

Features

- Update note title.
- Update note content.
- Restrict updates to note owners only.

Deleting Notes

Users can permanently remove notes.

Features

- Delete selected notes.
- Restrict deletion to note owners only.
- Remove records from the database.

Input Validation

Validation is implemented to ensure data integrity.

User Validation

- Username is required.
- Username must be unique.
- Password is required.
- Password must meet minimum security requirements.

Note Validation

- Title cannot be empty.
- Content cannot be empty.

Benefits

- Improves data quality.
- Prevents invalid records.
- Enhances application reliability.

Error Handling

The API returns meaningful responses for invalid operations.

Examples

- Duplicate username detection.
- Invalid login credentials.
- Unauthorized access attempts.
- Note not found scenarios.

HTTP Status Codes Used

200 OK

Request processed successfully.

201 Created

Resource created successfully.

400 Bad Request

Invalid request data.

401 Unauthorized

Authentication token missing or invalid.

404 Not Found

Requested resource does not exist.

500 Internal Server Error

Unexpected server error.

Project Output

The completed application successfully achieved the following results:

- Users can register securely with encrypted passwords.
- Users can log in using valid credentials.
- JWT tokens are generated for authenticated users.
- Protected endpoints can only be accessed by authorized users.
- Users can create personal notes.
- Users can view only their own notes.
- Users can update existing notes.
- Users can delete notes when no longer needed.
- Input validation prevents invalid data entry.
- Error handling provides meaningful responses.

Result

The Secure Note-Taking API successfully provides a secure and reliable note management system using ASP.NET Core Web API. The application implements authentication, authorization, password hashing,

validation, and CRUD operations while ensuring that user data remains protected and accessible only to authorized users.